ECHOTALK: MASTER PRODUCTION & ARCHITECTURE SPECIFICATION

INSTRUCTIONS FOR AI GENERATOR / GOOGLE FLOW: Execute the full production build for **EchoTalk** based on the structured specifications below. Build a zero-lag, modern borderless social app with advanced media tools, Gemini AI integration, high-production arcade games, in-DM voice/video calls, dynamic audio/haptics, fluid auto-adjusting layouts, an automated moderation vault, and an autonomous self-healing code architecture.

1. Universal Zero-Lag Performance & Auto-Healing Core

- **0ms Perceived Input Latency:** Apply touch-action: manipulation globally to eliminate 300ms touch delays.
- **GPU Hardware Compositing:** Force GPU hardware compositing across all swipeable views, chat streams, game boards, and modals using transform: translate3d(0,0,0) and will-change: transform, opacity.
- **60–120 FPS Rendering:** Enforce requestAnimationFrame and OffscreenCanvas for interactive components (photo editor, doodle canvas, game boards).
- **Virtualized Lists:** Implement virtualized list windowing for DM chat streams, Discover lists, and Lounge member directories to guarantee zero scrolling lag regardless of DOM size.
- **Autonomous Self-Healing System:**
    - Wrap every tab, modal, game canvas, and message stream in localized React Error Boundaries.
    - If a runtime error occurs, silently reset local state and re-render without crashing or displaying a blank screen.
    - Auto-repair corrupted JSON user settings on app load by restoring system defaults instantly.
    - Queue offline writes (IndexedDB / local state) during network drops and re-sync state seamlessly once connection is restored.

2. Fully Dynamic Auto-Adjustable Layout Engine

- **Universal Auto-Layout Rules:**
    - Do NOT use hardcoded pixel heights or rigid aspect ratios (height: 800px or height: 100vh) on main screens or chat threads.
    - Enforce dynamic viewport units (100dvh) and flexible containers (flex: 1 1 auto; min-height: 0;).
- **DM Chat Auto-Calculating Layout:**
    - Main chat container uses display: flex; flex-direction: column; height: 100dvh; max-height: 100dvh; overflow: hidden;.
    - Message history stream uses flex: 1 1 auto; min-height: 0; overflow-y: auto;

padding-bottom: 12px; to automatically shrink or expand and push the input bar up/down.
    - Bottom input bar uses flex: 0 0 auto; height: auto; padding-bottom: max(12px, env(safe-area-inset-bottom)); so it expands dynamically when typing multi-line text and sits naturally above the screen boundary.
- **Profile Screen Stacking Layout:**
    - Photo carousel uses an auto-fitting container (height: clamp(300px, 42vh, 480px); width: 100%; position: relative;) with object-fit: cover;.
    - Pagination dots position absolutely at bottom: 12px inside the photo container with z-index: 10; pointer-events: none;.
    - Action Bar (Like, Message, Friend, Gift) sits in the normal flex layout **below** the photo slider with position: relative; z-index: 20; display: flex; flex-wrap: nowrap; overflow-x: auto; padding: 12px 16px; gap: 10px;. Enable smooth touch scrolling (scroll-snap-type: x mandatory; touch-action: pan-x;) if buttons exceed screen width.
    - User Details Card takes height: auto; width: 100%; and flows naturally down the scrollable container.

3. Borderless Adaptive Design System & 5 Themes

- **Borderless Aesthetics:** Remove heavy borders and boxy outlines in favor of elevation shadows, soft surface color shifts, and dynamic backdrop blurs (backdrop-filter: blur(20px)).
- **5 Unified Presets:**
    1. **Classic (Default Light):** Off-white background (#F8F9FA), soft elevated cards (#FFFFFF), dark charcoal text (#111827), sapphire accents (#2563EB).
    2. **iOS Glassmorphism (Dark Translucent):** Deep navy/black (#0B0F17), translucent glass panels (rgba(255,255,255,0.06)), frosted blur, neon indigo accents (#6366F1).
    3. **Warm Sand & Cream (Fresh Soft):** Cream/beige background (#FDFBF7), soft skin-tone cards (#F5EBE0), deep coffee text (#3D2C2E).
    4. **Midnight Emerald (Rich Dark):** Forest charcoal background (#0D1B1E), elevated teal cards (#162A2D), gold/mint accents (#10B981).
    5. **Obsidian Pure Dark (OLED Black):** Pitch black (#000000), dark graphite cards (#121212), neon violet accents (#8B5CF6).
- **Adaptive Contrast System:** Typography, icons, and vectors automatically recalculate contrast based on theme selection so white icons/text never vanish against light backgrounds.

4. Permission-Gated Audio/Video Calling & Screen Sharing

- **Mutual Acceptance Gate:** Audio Call (📞) and Video Call (📹) icons in DMs are **strictly disabled/hidden until both users accept a mutual friend request**.
- **Full-Screen Call View:** WebRTC audio and video calling interface featuring Mute Mic, Camera Flip, Floating Self-View, and End Call controls.
- **Live Screen Sharing:** Include a **Screen Share (🖥️)** control inside active video calls to

broadcast mobile screen content live in real time.

5. View-Once Self-Destructing Media & Telegram Particle Dissolve

- **Flame View-Once Toggle (🔥):** Add a Flame icon toggle in the DM media picker. When active, sent photos or doodles can only be viewed **once** by the recipient before automatically self-destructing.
- **Universal Particle Dissolve:** Sent/received text, voice notes, photos, and doodles in DMs disintegrate into a Telegram-style 400\text{ms} pixel-dust particle animation upon deletion ('Delete for Me' / 'Delete for Everyone').

6. Arcade Hub: Zero-Delay Touch-and-Hold Peek Engine

- **Clean Logo Grid:** Minimalist icon-only logos for **Chess**, **Ludo**, and **Mafia** without card backgrounds or subtext.
- **Immediate Touch-and-Hold Gesture:** On onTouchStart, immediately scale the logo (transform: scale(1.05)) with spring physics and open a frosted glass info modal showing rules, player count, and 'Play Now'.
- **Instant Release Snap-Back:** Releasing touch (onTouchEnd) snaps the preview closed with zero delay.
- **Quick Tap Launch:** Single quick tap under 200\text{ms} opens the full-screen game directly.

7. Dynamic Audio, Custom Haptics & Soundscapes

- **Tactile Haptic Engine (navigator.vibrate):**
  - Light 10\text{ms} tap for tab switching.
  - Double-pulse for DM emoji reaction pops.
  - Heavy bump when a Ludo token lands on base or a Chess piece captures.
  - Continuous micro-vibration along the Blur Brush path in the photo editor.
- **Ambient Soundscapes in Lounge:** Optional toggleable background audio (lo-fi chill, cafe rain, neon synthwave) in Lounge voice and public chat rooms.

8. High-Production Arcade Games Suite

- **Ludo Master:** 15x15 board with dynamic 3D rolling dice box that glides and glows in the active player's corner base (Red, Green, Yellow, Blue).
- **Roundtable Mafia:** Circular avatar layout with live speaker rings, smooth Day/Night phase lighting shifts, private swipe-up role cards, and live voting tallies.
- **Chess Master:** Drag-and-drop spring physics, legal move highlights, Castling, En Passant, Pawn Promotion, Web Audio API sound FX, and AI / 1v1 DM play.
- **In-Game Voice & Spectator Mode:**
  - Floating in-game voice bar on Chess, Ludo, and Mafia boards for live talk during

matches.
    - Spectator Mode allowing users to watch ongoing friend matches and fire live animated emoji taunts across the board.

9. Google Pixel DM Context Menu, Media Editor & AI Power

- **Long-Press Message Context Menu:** Long-pressing any DM bubble displays a floating glass bar:
    - Top Row: Quick emoji reactions (❤️, 😂, 🔥, 👍, 😮, 😢) attaching animated badges to message corners.
    - Action List: Copy Text, Translate (via Gemini API), and Delete (triggers particle dissolve).
- **Telegram-Grade Photo Editor:** Full-screen canvas with Crop, Text Markup, Neon Glow Brushes, and a dedicated **Blur Brush** (ctx.filter = 'blur(12px)') to redact sensitive details before sending.
- **Gemini AI Features:**
    - **AI Smart Reply Chips:** Non-intrusive 2-word quick-reply chips above the DM keyboard.
    - **Voice-to-Text Tone Polish:** Hold-to-record voice notes transcribed via Gemini AI with option chips to rephrase tone (*Casual, Formal, Poetic, Sarcastic*) before sending as text.
    - **One-Tap Page Translation (🌐):** Header switch in DMs routing visible messages through Gemini AI for live page-wide translation.

10. Profile Customization & Discover Filters

- **Custom Profile Banners:** Full-screen profile header supporting animated video/GIF banners and an optional 5-second favorite music preview playing softly when a profile is opened.
- **Custom Chat Bubble Accents:** Per-DM customization allowing users to set unique color gradient tints (e.g., Sunset Pink, Electric Cyan, Emerald Glow, Midnight Purple) for message bubbles.
- **Discover Filters:** Real-time multi-filter engine for **Gender** (All, Male, Female), **Age Range**, and **Country** with locked IP Verified flags.

11. Automated Content Scanning & Unencrypted Admin Quarantine Vault

- **Server-Side Content Scanning:** Run automated inspection on incoming text, photos, doodles, and **View-Once (Flame 🔥) media** before/during delivery.
- **Direct Raw Quarantine Pipeline:**
    - If content is flagged as explicit or unsafe, immediately store the file in its **raw, original format** (images as .jpg/.png, text as raw string/JSON) into the /admin-quarantine-folder/.

    - Do NOT apply client-side encryption or obscure file transformations.
    - View-Once media must be captured and saved in its original format at transmission before self-destruction occurs on the recipient device.
- **Zero Cloud-Level Blocking Override:** Do not trigger hard cloud API blocks; handle all redirection smoothly at the app logic layer into the owner storage bucket.
- **Owner Admin Command Center:**
    - Exclusively render a hidden 'Owner Command Center' inside Settings for Owner role accounts.
    - Features a **Quarantine Vault** tab displaying raw thumbnails and text for single-tap review, user report logs, account warning/ban controls, and global feature toggles.

12. Critical Anti-Patterns ("What NOT To Do")

1. **DO NOT** hardcode fixed pixel heights (height: 800px or height: 100vh) on main screens or chat threads. Use dynamic viewport units (100dvh) and flexbox/grid containers (flex: 1 1 auto; min-height: 0;).
2. **DO NOT** use heavy border outlines (border: 1px solid #333), thick dividers, or boxy card contours. Rely on backdrop blurs, surface color shifts, and elevation shadows.
3. **DO NOT** allow profile photo sliders to position over active UI controls like the Action Bar or navigation dots.
4. **DO NOT** hide or clip interactive buttons on narrow screens. Enforce auto-wrapping or smooth horizontal touch scrolling (touch-action: pan-x).
5. **DO NOT** add artificial delay timers for Arcade peeks. Trigger peeks on immediate touch (onTouchStart).
6. **DO NOT** leave standard 300ms mobile touch delays active on buttons. Apply touch-action: manipulation globally.
7. **DO NOT** lock or hide the bottom DM text bar behind the screen edge or system gesture bars.
8. **DO NOT** allow audio/video call controls to be accessible in DMs before a mutual friend request is explicitly accepted.
9. **DO NOT** perform un-accelerated CSS animations. Never animate width, height, top, or left—use GPU-composited properties (transform: translate3d(...) and opacity) exclusively.
10. **DO NOT** allow component errors or API timeouts to trigger a white screen or crash the app. Always catch errors gracefully in React Error Boundaries.